

Long-Short Term Memory (LSTM)

The Problem of Long-Term Dependencies

- **Key Characteristics of RNN (Recurrent Neural Network)**
- Processes **sequential data** effectively.
- Maintains a **hidden state (memory)** to store previous information.
- Uses **feedback/recurrent connections**.
- Captures **temporal dependencies** in data.
- Considers both **current input and previous hidden state** when generating output.
- Suitable for tasks involving **contextual information**.
- Commonly used in **Natural Language Processing (NLP)**, speech recognition, and time-series forecasting.

Drawbacks of Recurrent Neural Networks (RNNs)

- **1. Vanishing Gradient Problem**
- During training, RNNs use **Backpropagation Through Time (BPTT)** to update weights. As the error is propagated backward through many time steps, the gradients can become extremely small (approach zero).
- **Effect:**
- Earlier layers receive almost no learning signal.
- The network fails to learn relationships between distant inputs.
- **Example:**
- Consider the sentence:
- "I grew up in France. ... I speak fluent French."
- To predict the word **French**, the network must remember **France** from many words earlier. Due to vanishing gradients, a standard RNN may forget this information.

Drawbacks of Recurrent Neural Networks (RNNs)

- **2. Exploding Gradient Problem**
- Sometimes gradients become extremely large during backpropagation.
- **Effect:**
- Weight updates become excessively large.
- Training becomes unstable.
- The loss function may fluctuate dramatically
- **Example:**
- If a gradient becomes 1000 instead of 0.5, the weight update will be enormous, causing the model to overshoot the optimal solution.
- **Solution:**
- LSTM and GRU architectures

Drawbacks of Recurrent Neural Networks (RNNs)

- **3. Difficulty Learning Long-Term Dependencies**
- Standard RNNs are effective at learning short-term patterns but struggle with information that occurred many time steps earlier.
- **Effect:**
- Important historical information is forgotten.
- Performance decreases for long sequences.
- **Example:**
- In language translation, a word appearing at the beginning of a sentence may influence words at the end. RNNs often fail to preserve this information.
- **Why?**
- The hidden state continuously gets updated, causing older information to gradually disappear.

Drawbacks of Recurrent Neural Networks (RNNs)

- **Limited Memory Capacity**
- Although RNNs have a hidden state that acts as memory, this memory is limited.
- **Effect:**
- Only recent information is effectively retained.
- Older information tends to fade away.
- **Example:**
- In a 500-word document, an RNN may remember information from the last few words but struggle to recall information from the beginning.

Long Short-Term Memory

- **LSTM** stands for **Long Short-Term Memory**, a type of recurrent neural network (**RNN**) used in machine learning for processing sequences of data.
- It was designed to solve a problem with traditional RNNs: they struggle to remember information from far back in a sequence because of the **vanishing gradient problem**.

How LSTM Works

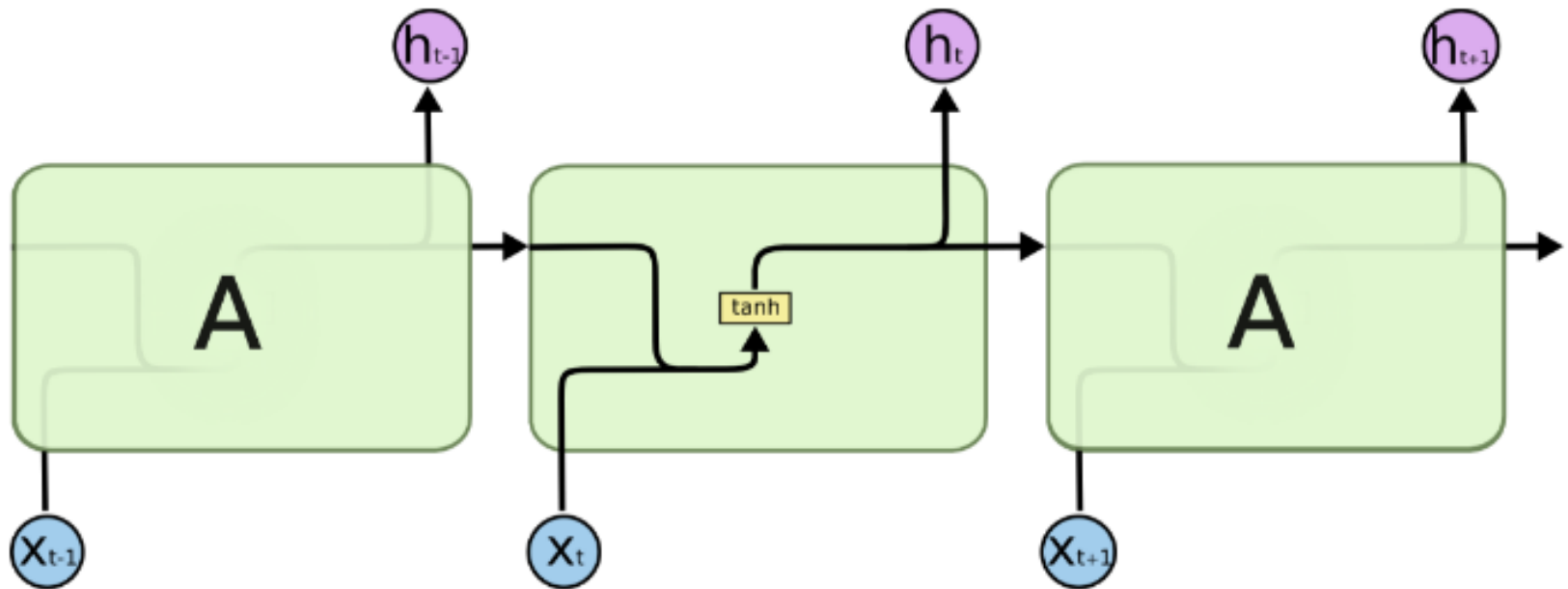
- An LSTM has a special memory cell and three main gates that control information flow:
- **Forget Gate** – Decides what information to discard from memory.
- **Input Gate** – Decides what new information to store.
- **Output Gate** – Decides what information to output.
- These gates allow the network to keep important information for long periods and forget irrelevant details.

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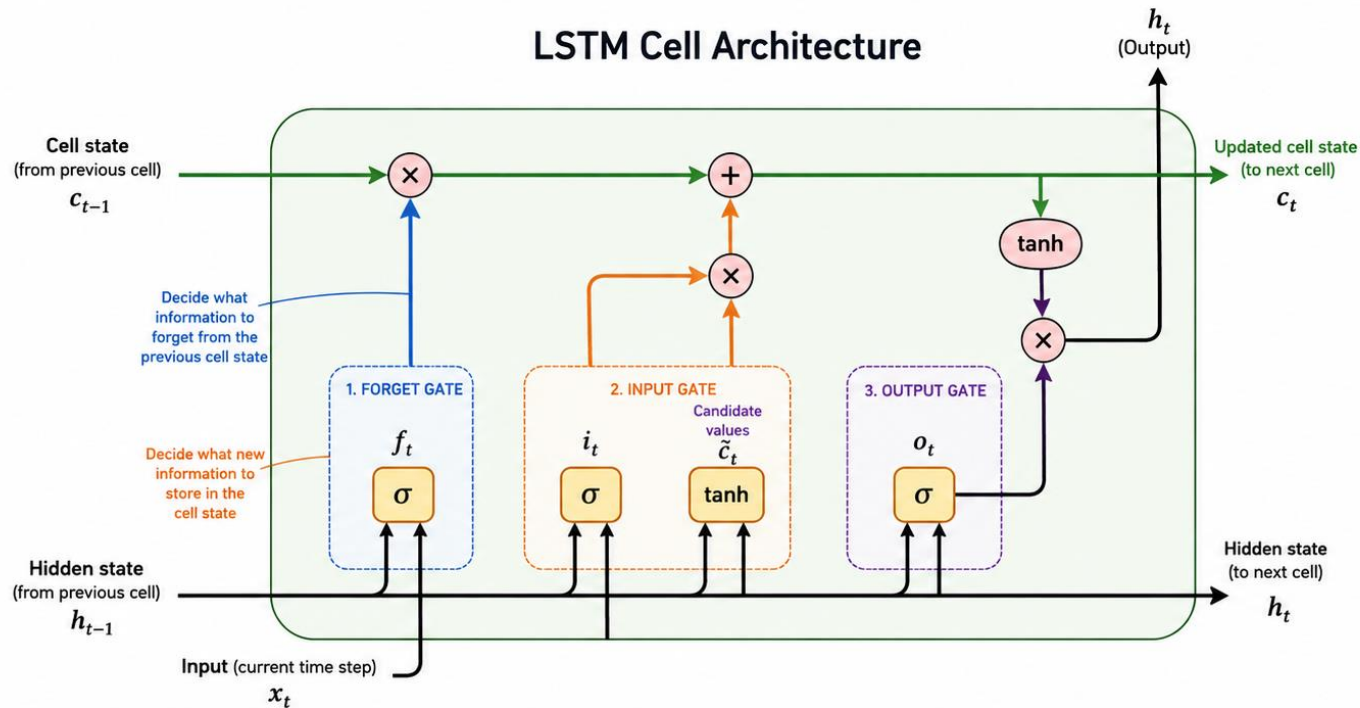
Understanding LSTM Architecture

Image is taken from <http://colah.github.io/posts/2015-08-Understanding-LSTMs/>



Repeated blocks in RNNs

LSTM Cell Architecture



1. FORGET GATE

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

Outputs a number between 0 and 1 for each number in the cell state.

1 = keep completely
0 = forget completely

2. INPUT GATE

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$

Decides which values to update (i_t) and creates new candidate values ($\tilde{c}_t$).

3. OUTPUT GATE

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

Decides which part of the cell state should be output as hidden state.

CELL STATE UPDATE

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

Where,

f_t = forget gate

i_t = input gate

$\tilde{c}_t$ = candidate values

HIDDEN STATE (OUTPUT)

$$h_t = o_t \odot \tanh(c_t)$$

Where,

o_t = output gate

$\odot$ = element-wise multiplication

LEGEND

σ Sigmoid activation (0 to 1)

$\tanh$ Tanh activation (-1 to 1)

$\otimes$ Point-wise multiplication

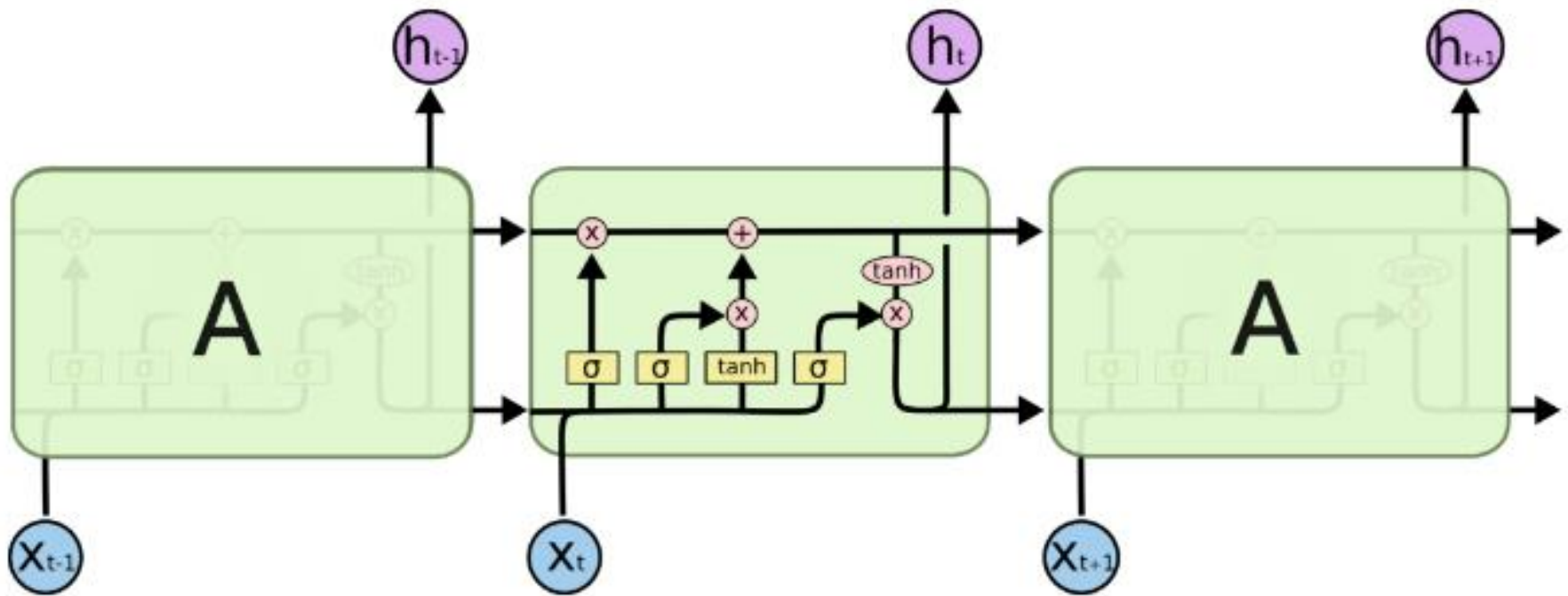
$+$ Point-wise addition

$\rightarrow$ Cell state flow

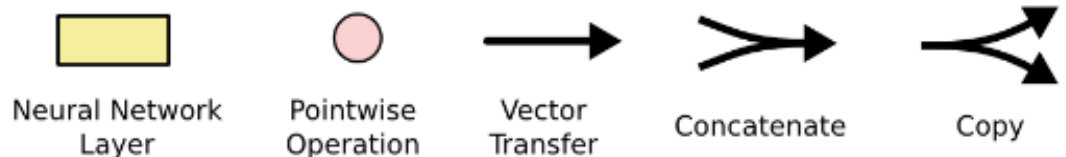
$\rightarrow$ Hidden/Input flow

Understanding LSTM Architecture

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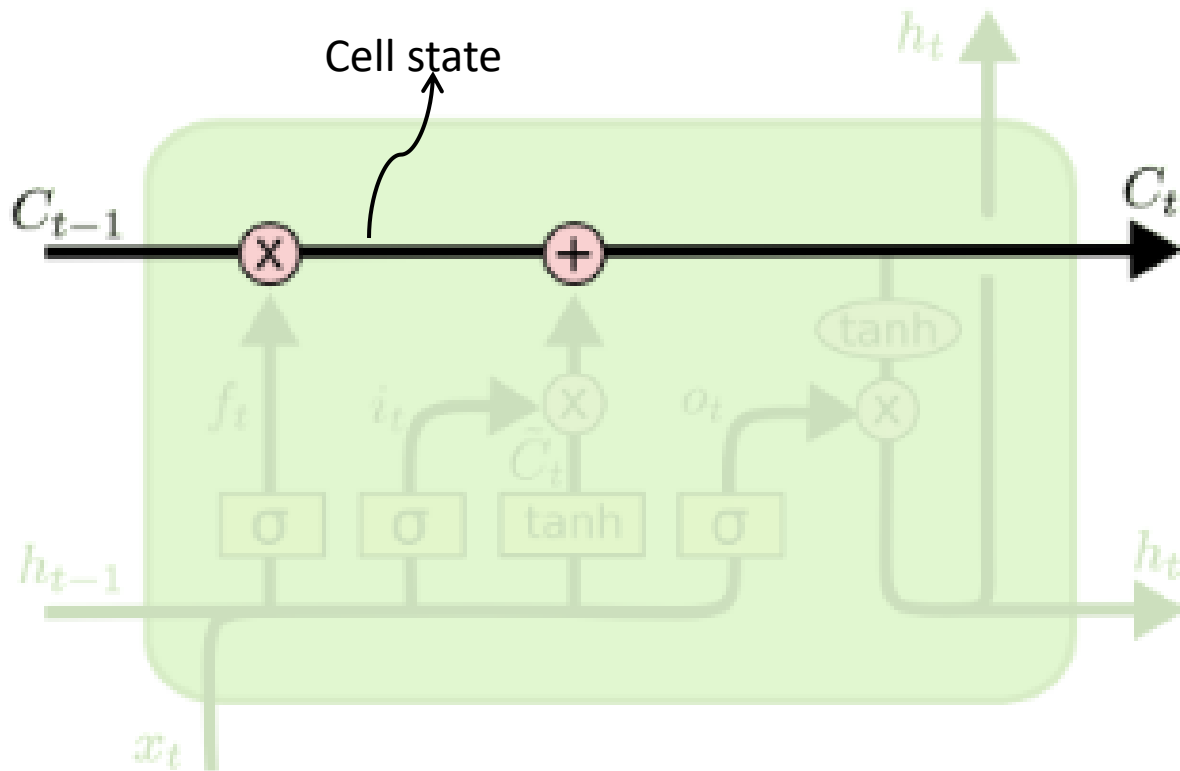


Repeated blocks in LSTMs



Understanding LSTM Architecture

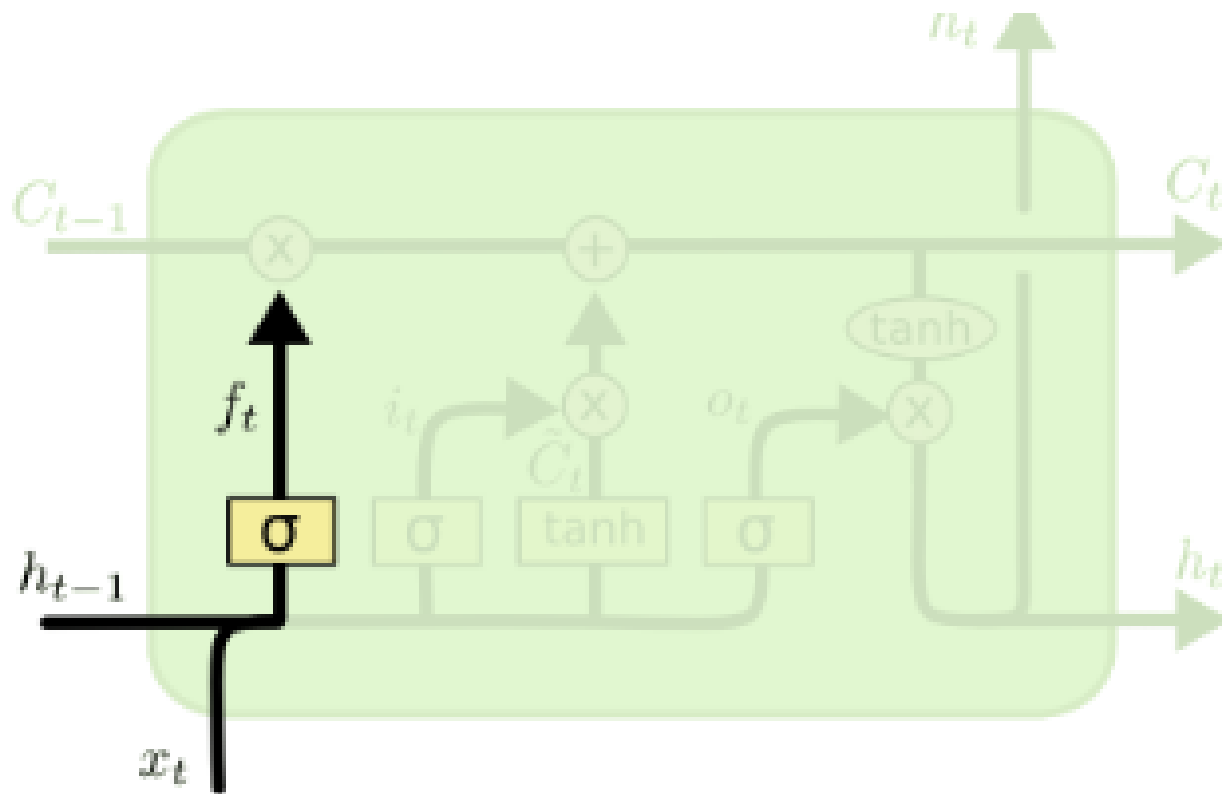
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A unit of LSTMs

Understanding LSTM Architecture

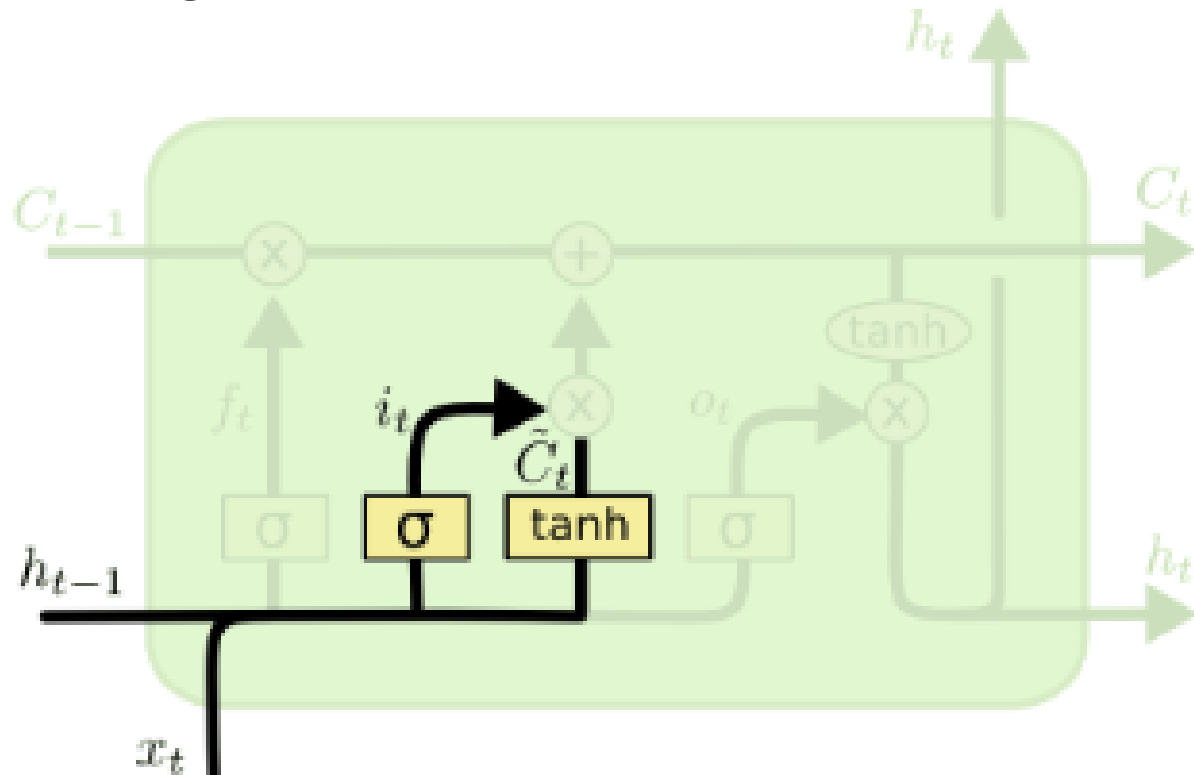
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$$f_t = \sigma (W_f \cdot [h_{t-1}, x_t] + b_f)$$

Understanding LSTM Architecture

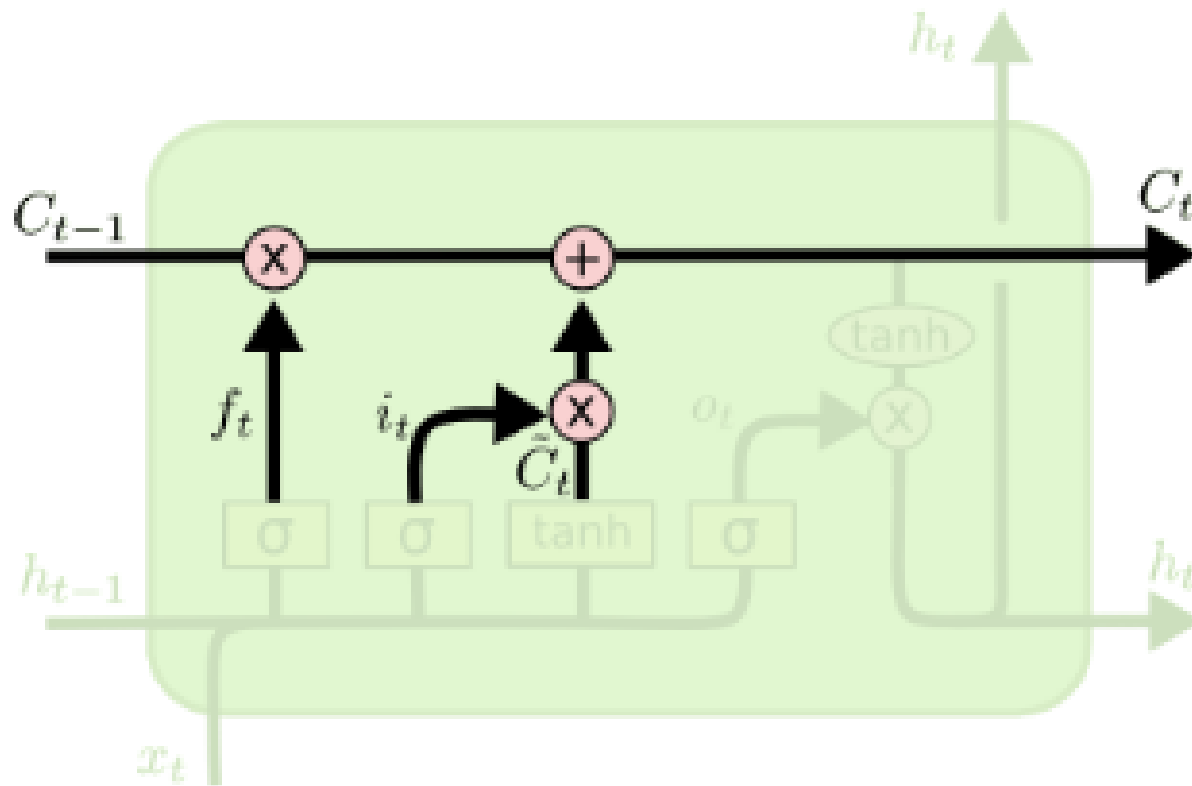
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$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$
$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

Understanding LSTM Architecture

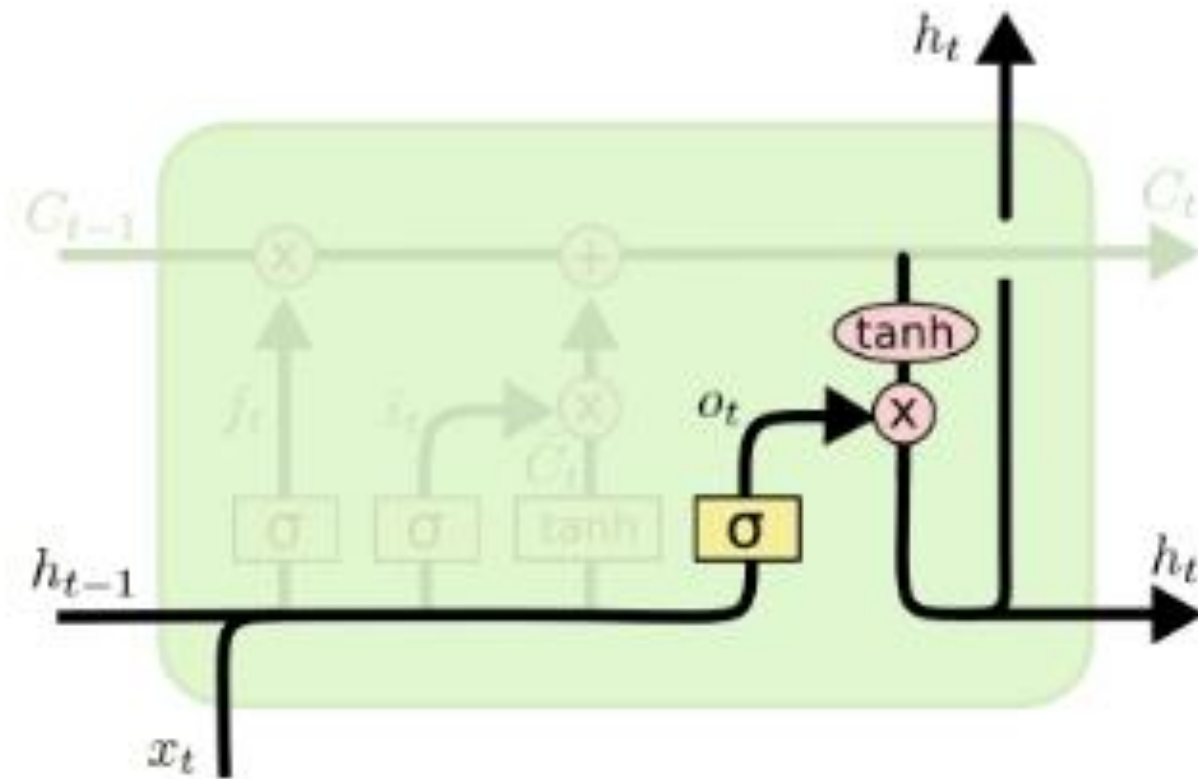
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$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

Understanding LSTM Architecture

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$$o_t = \sigma(W_o [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t * \tanh(C_t)$$

Component	Type	Main Purpose	Memory Type
C_t	Cell State	Stores long-term memory	Long-term
h_t	Hidden State	Output of LSTM	Short-term
f_t	Forget Gate	Removes old info	Control
i_t	Input Gate	Adds new info	Control
$\tilde{C}_t$	Candidate	New memory proposal	Temporary
o_t	Output Gate	Controls output	Control

Example of LSTM

- Consider a single LSTM cell at time step t .
- **Inputs:**
- Input $x_t = 1$
- Previous hidden state $h_{t-1} = 0.5$
- Previous cell state $C_{t-1} = 0.2$

Example of LSTM

- **Weights:**
- Forget gate: $W_f = 0.8$, $U_f = 0.4$, $b_f = 0.1$
- Input gate: $W_i = 0.7$, $U_i = 0.3$, $b_i = 0.2$
- Candidate gate: $W_c = 0.6$, $U_c = 0.2$, $b_c = 0.0$
- Output gate: $W_o = 0.9$, $U_o = 0.5$, $b_o = 0.1$

Example of LSTM

- **Activations (given):**
- $\sigma(x) \approx 0.7$
- $\tanh(x) \approx 0.6$

Example of LSTM

- **Task:**
- Compute:
- Forget gate f_t
- Input gate i_t
- Candidate cell state $\tilde{C}_t$
- New cell state C_t
- Output gate o_t
- Hidden state h_t

1. Forget Gate

- **Formula:**
- $f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$
- **Substitution:**
- $f_t = \sigma(0.8 \cdot 1 + 0.4 \cdot 0.5 + 0.1)$
- $f_t = \sigma(0.8 + 0.2 + 0.1) = \sigma(1.1)$
- **Value:**
- $f_t = 0.7$

2. Input Gate

- **Formula:**
- $i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$
- **Substitution:**
- $i_t = \sigma(0.7 \cdot 1 + 0.3 \cdot 0.5 + 0.2)$
- $i_t = \sigma(0.7 + 0.15 + 0.2) = \sigma(1.05)$
- **Value:**
- $i_t = 0.7$

3. Candidate Cell State

- **Formula:**
- $\tilde{C}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c)$
- **Substitution:**
- $\tilde{C}_t = \tanh(0.6 \cdot 1 + 0.2 \cdot 0.5 + 0)$
- $\tilde{C}_t = \tanh(0.6 + 0.1) = \tanh(0.7)$
- **Value:**
- $\tilde{C}_t = 0.6$

4. New Cell State

- **Formula:**
- $C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$
- **Substitution:**
- $C_t = (0.7 \cdot 0.2) + (0.7 \cdot 0.6)$
- $C_t = 0.14 + 0.42$
- **Value:**
- $C_t = 0.56$

5. Output Gate

- **Formula:**
- $o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o)$
- **Substitution:**
- $o_t = \sigma(0.9 \cdot 1 + 0.5 \cdot 0.5 + 0.1)$
- $o_t = \sigma(0.9 + 0.25 + 0.1) = \sigma(1.25)$
- **Value:**
- $o_t = 0.7$

6. Hidden State

- **Formula:**
- $h_t = o_t \cdot \tanh(C_t)$
- **Substitution:**
- $h_t = 0.7 \cdot \tanh(0.56)$
- **Given:**
- $\tanh(0.56) \approx 0.6$
- **Value:**
- $h_t = 0.7 \cdot 0.6 = 0.42$

- **ANSWERS**

- $f_t = 0.7$

- $i_t = 0.7$

- $\tilde{C}_t = 0.6$

- $C_t = 0.56$

- $o_t = 0.7$

- $h_t = 0.42$